CFD Simulation and Analysis of Oil-Water Separation in a Horizontal Gravity Separator

Fluent - Conferences et cas d'etude

Taha JAVED, Busayomi David THOMPSON-AJAYI

December 30, 2024

Contents

1 Introduction **2**

2 The Studied Problem **2**

3 Setup of the Problem **4**

4 Results and Discussion 4

5 Parametric Analysis **7**

6 Conclusion: **8**

7 References: **8**

1 Introduction

This report investigates multiphase flows in oil water separation, how oil interacts with water, the behaviour of both fluids when they enter the separation chamber and how both the fluids behave under the influence of gravity. Oil-water separation is a critical process in industries such as petroleum refining, wastewater treatment, and marine engineering. Effective separation ensures environmental compliance and reduces operational costs. The technological aim is to better understand and quantify the behavior of an oil–water mixture in a horizontal gravity separator through CFD simulations. Rather than focusing on full design optimization, this study aims to identify key factors that influence separation efficiency (such as flow rates, geometry, and fluid properties) and provide basic guidelines for improving operational performance in industrial applications. The scientific aim is to explore the fundamental multiphase flow principles governing oil–water separation under the influence of gravity, flow velocity, and turbulence in a horizontal separator. The study examines how the laminar, k-epsilon and k-omega models affect predictions of phase distribution, pressure, and velocity fields, thereby offering insights into the most appropriate modeling approaches for accurately capturing oil–water flow dynamics.

2 The Studied Problem

The primary problem studied is the separation of oil and water using a gravity-based separator. The separator's performance depends on parameters such as flow rate, droplet size, and turbulence levels. The geometry consists of an inlet chamber, separation chamber, and outlet chamber. The flow dynamics are modeled using ANSYS Fluent to predict velocity profiles, pressure distribution, and phase separation efficiency.

The fluids are oil (density: 870 kg/m³, viscosity: 0.01 Pa·s) and water (density: 998.2 kg/m³, viscosity: 0.001 Pa·s). The oil–water surface tension is taken as 0.03 N/m. The gravity direction is vertical, downward and the domain is of length 2.5 m and height 0.5m as shown in figure 1.

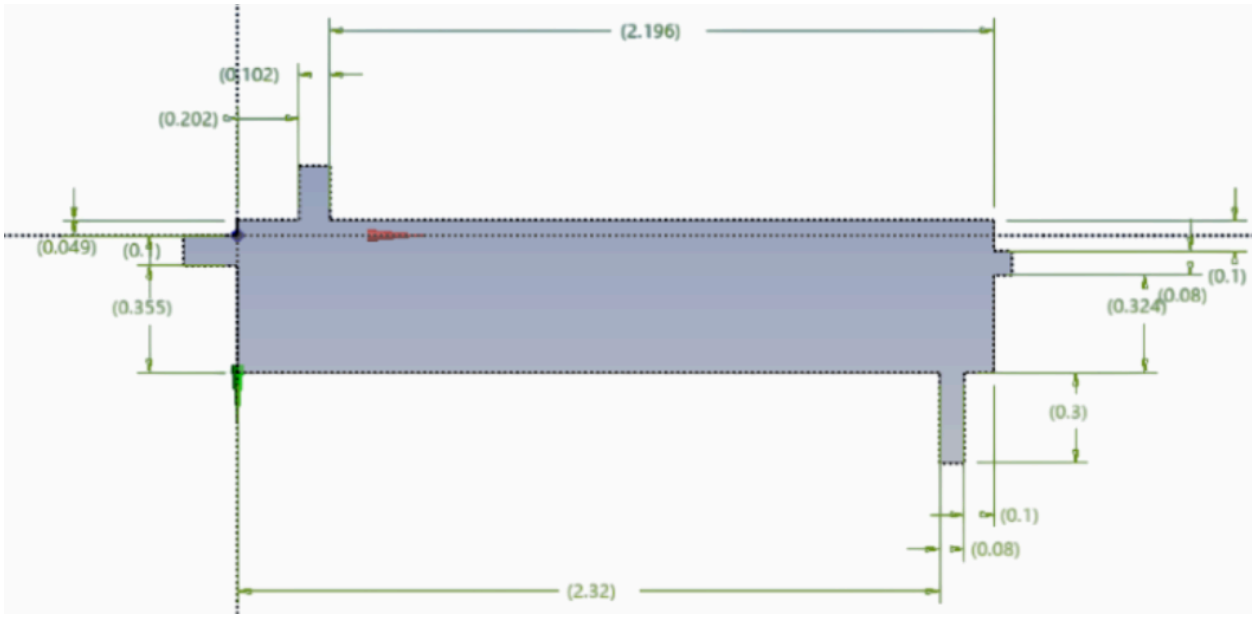


Figure 1 : Geometrical Sketch

3 Setup of the Problem

Numerical Approach

CFD Code: ANSYS Fluent.

Boundary Conditions:

Inlet Velocity (Water): 1m/s

Inlet Velocity (Oil) : 1m/s

Walls: No-slip condition.

Gravity: Enabled.

Mesh Configuration

We can see the mesh in the figure 2.



Figure 2 : Mesh Image (Refined)

Mesh type: Structured mesh.

Total Cells : 82802 cells

Initial Condition

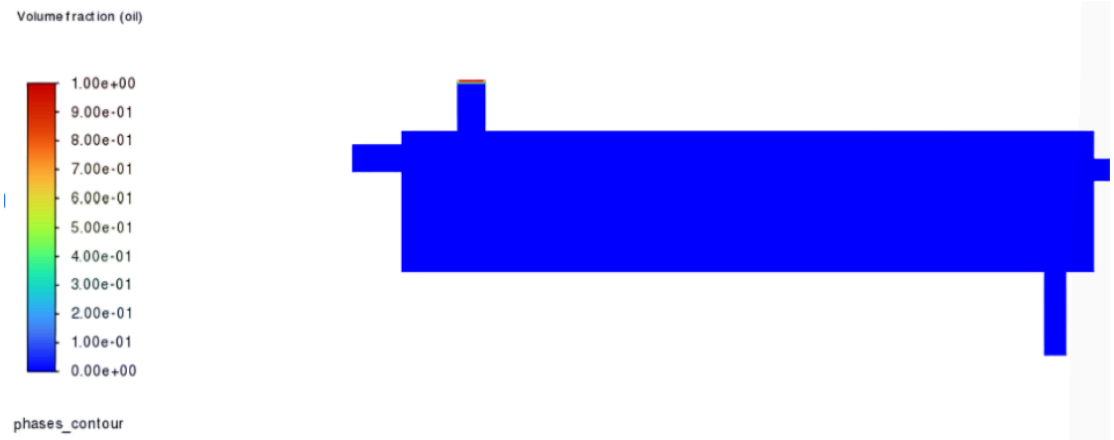


Figure 3 : Initial Condition when the chamber is filled with only water

The separator initially contains only water.

4 Results and Discussion

Reynold's Number:

$$\mathbf{Re} = \frac{\rho * v * D}{\mu}$$

$\mathbf{Re_{oil}} = \frac{870 * 1 * 0.5}{0.01} = \mathbf{43{,}500}$ (turbulent flow) $\mathbf{Re_{water}} = \frac{998.2 * 1 * 0.5}{0.001003} = \mathbf{497{,}606}$ (turbulent flow)

The flow of both oil and water phases within the separator is characterized as turbulent, as indicated by their high Reynolds numbers, which exceed the critical threshold for turbulent flow. Turbulence plays a significant role in determining the flow dynamics and separation efficiency within the system, particularly in the interactions between the phases and near the walls of the separator.

Given the turbulent nature of the flow, the use of turbulence models such as k-ε and k-ω is both appropriate and essential for accurately capturing the behavior of the multiphase system. The k-ε model is well-suited for analyzing bulk flow turbulence and provides robust and computationally efficient solutions for the general flow field. On the other hand, the k-ω model offers improved accuracy in resolving near-wall effects, which are critical for understanding phase interactions and separation processes.

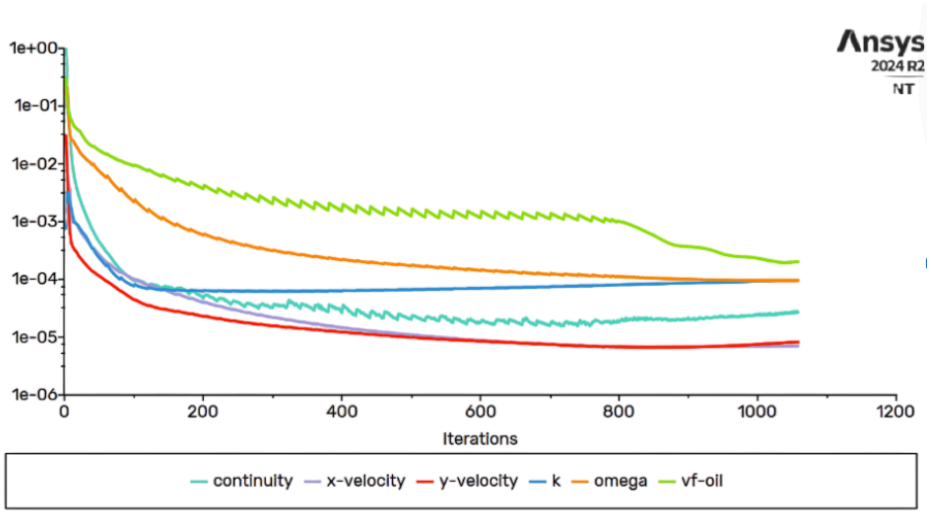


Figure 4: Residual plot

The figure 4 shows the residuals plot for an oil-water separator simulation in ANSYS Fluent using the fine mesh, representing the convergence behavior of key variables such as continuity, velocity components, turbulence parameters (k and omega), and volume fraction of oil (vf-oil) over iterations. The gradual decrease in residuals indicates that the simulation is approaching a numerically stable and converged solution, ensuring accurate phase separation analysis.

Mesh Convergence

Coarse: 868 cells /elements (0.04 size); 963 nodes

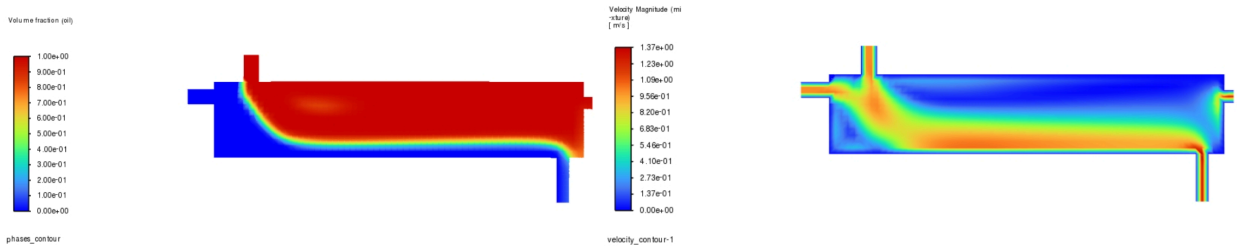


Figure 5 : Phase and Velocity Contours (Coarse)

The figure 5 shows coarse mesh that captures general flow behavior but lacks precision in resolving interface dynamics and velocity gradients, leading to less accurate phase separation results.

<u>Medium: Nodes: 13548 (size 0.01) ; Elements - 13175</u>

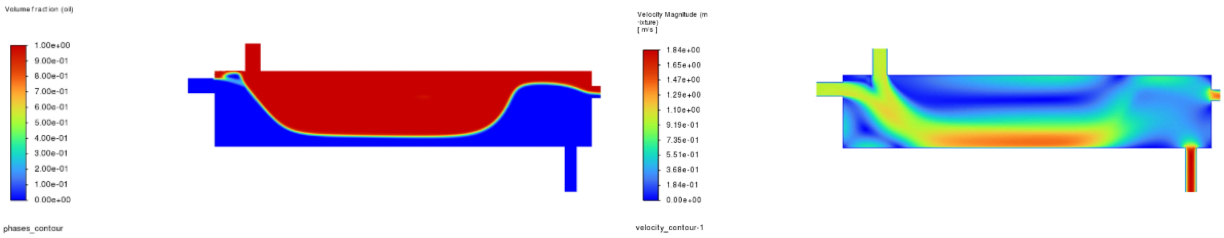


Figure 6 : Phase and Velocity Contours (Medium)

The figure 6 shows a medium mesh which improves resolution, showing clearer phase separation and velocity distribution, capturing key flow patterns and interface behavior more accurately.

<u>Fine: Nodes - 83731 (size = 0.004); (Elements = 82802)</u>

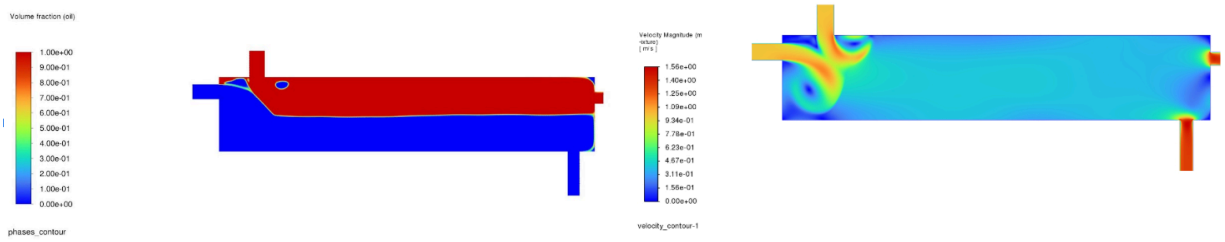


Figure 7 : Phase and Velocity Contours (Refined)

The figure 7 shows fine mesh which provides highly detailed flow characteristics, accurately resolving eddies, turbulence, and sharp interfaces, ensuring precise phase separation performance.

Velocity Profiles

Velocity profiles were analyzed to evaluate flow patterns and separation performance.

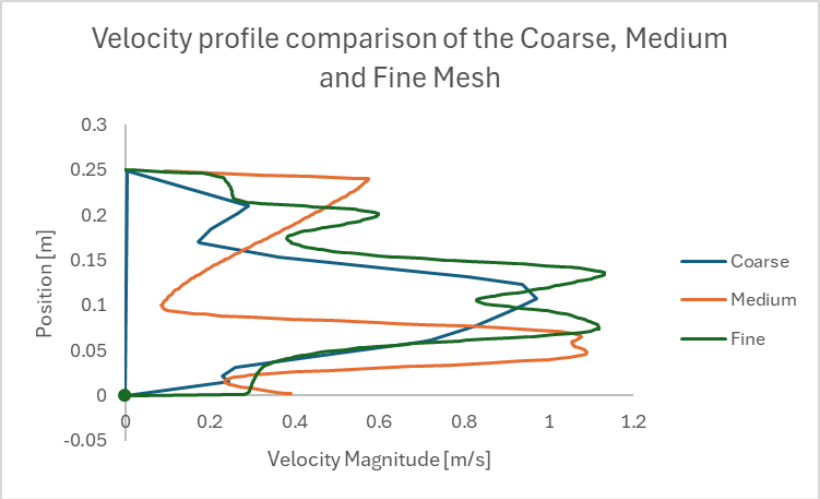


Figure 8 : Velocity profiles for Coarse, Medium and Fine Meshes

From figure 8 it is observed that the coarse mesh presents a very limited number of data points so the velocity profile is unable to properly capture the multiphase flow details in the separation chamber. The medium mesh provides a velocity profile with more data points and relatively more information on flow dynamics and behaviour. In the refined mesh, there are sufficient data points to capture single small details of fluid behaviour and provide a well defined velocity profile while retaining moderate computational cost.

5 Parametric Analysis

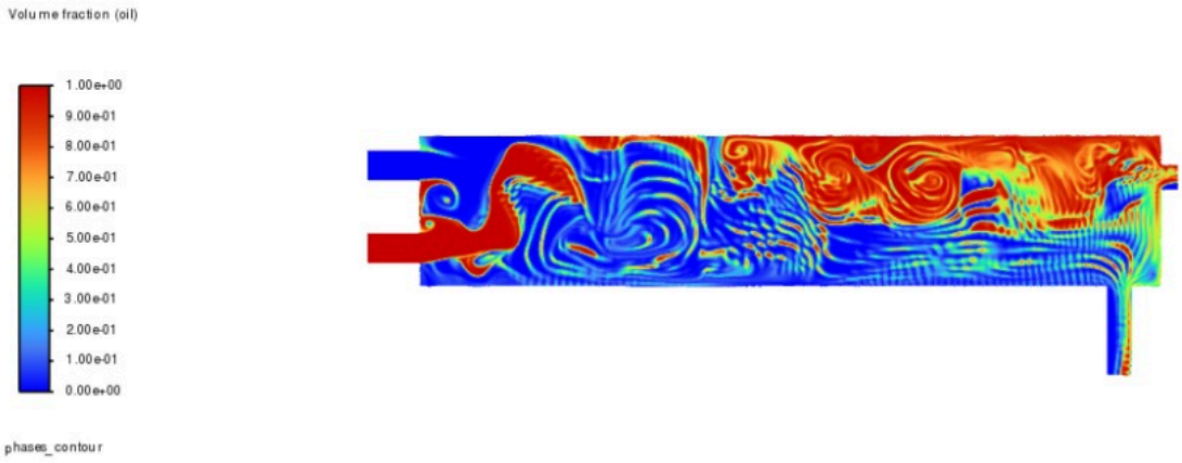


Figure 9 : Phase contour for changed oil inlet orientation (Laminar)

Here we changed the orientation of the oil inlet and here instead of the top inlet for oil we impose oil inlet from the extreme left side of the separation chamber in order to visualize more clearly the flow behaviour in oil water separator. For this scenario, due to the averaging done by the turbulent models and its approximation of small eddies, it was observed that the laminar case was best for simulating the flow of the water downwards and the rise of the oil phase due to density differences. From this, the inefficiency of the turbulent models in capturing heavy mixing was observed and the laminar model provided better results as shown in the figure 9.

6 Conclusion:

In this project, we gained valuable insights into model selection for multiphase flows. Initially, a laminar model was employed; however, it yielded unsatisfactory and sometimes divergent results. Consequently, the analysis shifted to turbulent models, including k–ε and k–ω, where k–ω proved especially effective for capturing near-wall phenomena and provided more stable, converged solutions. Additionally, optimizing the mesh to balance accuracy and computational cost presented a significant challenge. Nonetheless, the experience underscored the importance of aligning simulation parameters—such as turbulence models and grid resolution—with the flow characteristics under investigation.

7 References:

Wikipedia Contributors. *Oil–water separator*. Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/wiki/Oil–water_separator

Oil Skimmers, Inc.. *What is an Oil/Water Separator? How Does an Oil/Water Separator Work?*. https://www.oilskim.com/blog/how-does-an-oil-water-separator-work

Jonach, T.; Jordan, C.; Haddadi, B.; Harasek, M.. *Modeling and Simulation of 3-Phase Separators in the Oil and Gas Industry with Emphasis on Water Quality*. Chemical Engineering Transactions, 2022, 94, 1009-1014. https://www.cetjournal.it/cet/22/94/168.pdf.

Zhang, Y.; Li, J.; Wang, X.; Liu, Y.; Zhang, L.. *Numerical Simulation Study of Oil–Water Separation Based on a Super-Hydrophilic Copper Net*. Frontiers in Environmental Science, 2022. https://www.frontiersin.org/articles/10.3389/fenvs.2022.945192/full.

CFD Land. *Three-phase Oil-water-air Separator CFD Simulation, ANSYS Fluent Training*. https://cfdland.com/product/three-phase-oil-water-air-separator-cfd-simulation-ansys-fluent-training/